COMMERCIAL ITEM DESCRIPTION

DIVIDER, POWER, RADIO FREQUENCY

The General Services Administration has authorized the use of this Commercial Item Description.

Scope. This Commercial Item Description (CID), describes a wide band RF power splitter and reflectometer with calibrated line stretcher. It is operated for either transmission or reflection measurement with network analyzer. The unit was designed for use with the HP 11610B test port extension cable. This high quality semirigid coaxial cable is used with the network analyzer.

Classification. Equipment covered by this CID may be commercially available equipment modified to the extent necessary to meet the following description. The equipment shall be type III, class 5 style T and color R in accordance with MIL-T-28800.

Applicable documents.

Specifications and standards. The following specifications and standards form a part of this CID to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-T-28800	Test Equipment for use with Electronic Equipment, General Specification for.

(Copies of specifications, standards, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

Other publications. The following document forms a part of this CID to the extent specified herein. Unless otherwise specified, the issue of the documents which are DOD adopted shall be those liated in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI-C62.41-1980

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(Application for copies of ANSI publications should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

Order of precedence. In the event of a conflict between the text of th CID and the references cited herein (except for associated detail specifications, specification sheets or MS standards) the text of this description shall take precedence. Nothing in this CID however, shall supesede applicable laws and regulations unless a specific exemption has been obtained.

Salient characteristics. The equipment shall be capable of operation and be bid sample tested within the accuracies, limits, and specifications herein.

1. Safety. The equipment shall be UL (Underwriters Laboratories) listed and approved and/or shall comply with the type III safety requirements of MIL-T-28800.

a. Leakage current protection. Protection shall be provided from leakage current in excess of 3.5 milliamperes (mA) rms (AC and DC) from any accessible conductive parts of equipment (including control shafts with knobs broken or removed and recessed calibration or adjustment controls) to either pole of the power source for any position of the power switch(es).

b. Radio frequency interference (RFI) capacitors. Any capacitor used in RFI filters shall be rated for at least four(4) times the operating voltage.

c. Type III ventilation openings.

(1) Ventilation openings shall be designed to prevent accidental contact with live electrical circuits and mechanical moving parts.

(2) Top ventilization openings shall incorporate small mesh screening or similar design to prevent the expulsion of hot electronic particles due to failure of electronic devices such as capacitors.

2. Parts, materials and processes. Except for making, the divider, power, radio frequency shall comply with the type III parts, materials, and process requirements of MIL-T-28800. Nonstandard parts approval requests shall be in accordance with MIL-T-28800. Front panel marking shall be in accordance with MIL-T-28800.

a. Restricted material. The equipment shall comply with the restricted material requirements of MIL-T-28800, including the requirement for mercury and radioactive materials, and shall contain no combination of materials which cause deterioration of any material contained in the equipment due to the effects of outgassing.

3. Design and construction. Unless otherwise specified herein, the equipment shall comply with the type III design and construction requirements of MIL-T-28800. The equipment shall be of solid-state construction. Encapsulation and embedment (potting) of subassemblies may be used only for high voltage circuits where construction is susceptible to leakage or corona indicators and associated controls shall be in close proximity to each other,

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all potentiometer controls shall have at least a 270 degree rotation range (4.71 radians).

a. Type of construction. Construction of the reflection/transmission test unit shall be of a monolithic or a separate component design.

b. Modular design restrictions. The modular design concept for the reflection/transmission test unit is acceptable with the following restrictions;

(1) The divider, power, radio frequency shall consist of a mainframe and a single band plug-in.

(2) Each module shall contain power circuit protection fuse.

(3) Each modules plug-in card shall be keyed and mated with the back plane receptacle so that modules may not be randomly placed into the mainframe.

(4) Modules shall not be removable by access through the front or rear of the mainframe without the unaided hand.

(5) The module interface shall come with one(1) blank space cover and one(1) extended board.

(6) A package part number shall be assigned to the reflection/transmission test unit as a whole system. This system part number shall be marked on the mainframe identification plate and all plug-in(s).

(7) All module interface shall be done in the back of the mainframe.

c. Signal input and output connectors. Unless otherwise specified, all signal input and output connectors shall be type BNC female.

d. Internal cooling. The equipment shall be capable of operating continuously for twenty four(24) hours at 55°C. If forced air is used, it shall be as spcified in MIL-T-28800.

(1) Forced air cooling motor. DC brushless motors shall be utilized for the forced air cooling motor in the equipment.

(2) Air filter. Equipment utilizing a cooling scheme which exhausts air from the equipment through the cooling fan shall not require an externally accessible air filter.

(3) Thermal switch. A thermal switch shall be provided which will interrupt circuit power to the divider, power, radio frequency when the internal temperature rises to the degree of possible damage to the electronic components. A minimum of 10°C hysteresis drop from this tripping temperature must occur before circuit power is again applied.

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4. Electrical power sources. The equipment shall operate from nominal commercial, military, and shipboard power sources of 100, 110, 120, 200, 220 and 240 volts ac (±10%) at line frequencies of 50 Hz to 400 Hz.

a. Electrical power connectors. The equipment shall operate with a ground at the primary source on either side of the single phase input.

b. Power consumption. Maximum power consumption of all components combined shall not exceed 15W watts.

c. Fuses. Fuse(s) shall be provided, on the exterior rear panel for replacement without entering the enclosure. The fuse(s) shall automatically disconnect the equipment from the power source in the event of excessive current requirements.

d. Surge and transient voltage protection. The divider, power, radio frequency shall contain surge and transient voltage protection internal of the enclosure. The transient voltage protection circuit shall be so designed as to prevent electronic device damage within the system per type III solid state tolerance bands of MIL-T-28800, the surge voltage protection circuit shall conform to ANSI/IEEE standard C62.41-1980 The surge protection circuit shall drain excess power due to surges directly to the ground conductor for personnel protection. Normal operation of the divider, power, radio frequency may be inhibited and false operational signals may be generated under input voltage transients greater than ±20 percent and frequency transient conditions greater than ±5 percent from within the nominal type III steady-state tolerance bands of MIL-T-28800, damage (other than fuse failure) to the divider, power, radio frequency shall not occur during voltage transient conditions greater than ±20 percent and frequency transient conditions greater than ±5 percent within the maximum limits (30 percent voltage transients and 10 percent frequency transients) specified in MIL-T-28800. Normal operation of the divider, power, radio frequency shall be automatically resumed when the transient has passed and any blown fuses are replaced. Blown fuses shall be externally accessible. Testing shall be in accordance with MIL-T-28800.

5. Dimensions and weight. The size and weight of individual component shall be consistent with current commercial capabilities, refer to paragraphs 5.a and 5.b of this CID.

a. Equipment dimensions. The nominal dimensions per component for the divider, power, radio frequency should be approximately 5 inches height x 17 inches wide x 18 inches deep.

b. Equipment weight. The weight including all accessories (excluding manual) per component, shall not exceed 34 lbs.

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6. Enclosure requirements. Unless otherwise specified herein, the equipment shall comply with the style T enclosure requirements of MIL-T-28800.

a. Rackmount. Rackmount capability shall be available as an option for the equipment specified herein.

b. Tilt-bail handle. A tilt-bail handle shall be provided in lieu of the hinged metal handles in accordance with MIL-T-28800. Exception to this paragraph may be acceptable as this is not a prime consideration of the equipment.

c. Reusable pouch or container. An attached accessory pouch may be used for storage of all supplied accessories in lieu of front-cover storage.

7. Marking and identification.

a. Marking. The visible surface adjacent to panel facilities such as variable controls, indicators, jacks, connectors and fuse holder(s), shall be marked with a suitable word, phrase, or abbreviation thereof, indicating the purpose or use of the part.

(1) Panel marking. The front panel marking shall comply with the type III parts, materials and processes panel marking requirements of MIL-T-28800, however, marking identification processes other than the type III requirements specified in MIL-T-28800 may be used, provided the marking and identification processes are capable of passing the front panel marking test when tested in accordance with this CID.

b. Items to be marked (if applicable).

(1) Volume and weight (including accessories) shall be marked on three external surfaces of the enclosure.

(2) The type designation of the AN nomenclature of the Divider, Power, Radio Frequency shall be stenciled on the exterior or main frame on at least three adjacent surfaces, on the transient case or two opposite sides. Marking shall be in block letters approximately 2 inches in height.

8. Environmental requirements.

a. Temperature requirements. Unless otherwise specified herein, the equipment shall meet its specified performance characteristics and other requirements when subjected to the class 5 temperature requirements specified in MIL-T-28800 and as listed below.

(1) Non-operating temperature. The equipment shall meet its performance and accuracy requirements in an non-operating environment of -62°C to +85°C.

(2) Operating temperature. The equipment shall meet its performance and accuracy requirements in an operating environment in accordance with MIL-T-28800.

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b. Humidity requirements. The equipment shall meet its required performance and accuracy requirements under humidity requirements for class 5 equipment as specified in MIL-T-28800.

c. Altitude requirements. The equipment shall meet its required performance and accuracy requirements under the altitude requirements for class 5 equipment as specified in MIL-T-28800.

d. Vibration requirements. The equipment shall meet the class 5 vibration requirements as specified in MIL-T-28800.

e. Mechanical shock requirements.

(1) Shock, functional. The equipment shall meet the class 5 shock, pulse requirement as specified in MIL-T-28800.

(2) Transit drop. The equipment shall meet transit drop requirement as specified in MIL-T-28800.

(3) Bench handling. The equipment shall meet the requirements as specified in MIL-T-28800.

(4) Shock, High impact. The equipment shall meet requirements as specified in MIL-T-28800.

f. Electromagnetic interference (EMI). The equipment shall comply with MIL-T-28800.

g. Magnetic environment. The equipment shall comply with the magnetic environment requirements of MIL-T-28800.

9. Reliability. Reliability verification testing, shall be in accordance with the type III, class 5 environmental conditions of MIL-T-28800.

a. Mean time between failures (MTBF). The equipment mean time between failures (MTBF) shall be 2,000 hours.

b. Pre-conditioning requirement. Each deliverable equipment shall be capable of being subjected to the environmental screening effects of burn in and thermal cycling, with thermal cycling being accomplished in the sample testing phase.

(1) Burn-in. Burn-in shall expose the equipment to extended periods of elevated temperatures, (40°C for a minimum of 100 hours). Each equipment which fails during burn-in shall be repaired and fully retested, until failure free. Failures which occur during the burn-in test shall be noted and reported but shall not count toward the establishment of MTBF. Prior to and following burn-in, a satisfactory operation test (SOT) shall be performed. A satisfactory operation check (SOC) shall be conducted at selected times during the burn-in test.

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(2) Thermal cycling. Thermal cycling shall expose the equipment to hot and cold cycles for class 5 equipment per MIL-T-28800. Each equipment which fails during thermal cycling shall be repaired and full cycling test shall be noted and reported but shall not count toward the cycling, a satisfactory operation check (SOC) shall be conducted during the last two upper and lower temperature limits.

10. Calibration and maintenance adjustments. The design of the system shall provide for readily accessible calibration adjustments and maintenance adjustments. These adjustments shall be provided by variable value components which are adjustable by the use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules.

a. Maintainability. The equipment shall meet the type III maintainability requirements specified in MIL-T-28800.

11. Interchangeability. The equipment shall meet the type III Interchangeability requirements specified in MIL-T-28800.

12. Digital interface. The equipment shall meet the digital interface requirements per MIL-T-28800.

13. Performance characteristics.

Frequency range 2 to 12.4 GHz.

Impedance 50 ohm nominal.

Source reflection coefficient 0.09, 2.0 to 8.0 GHz, 0.13, 8.0 to 12.4 GHz, 0.2, 12.4 to 18 GHz.

Termination reflection coefficient 0.13 in reflection mode, 2.0 to 12.4 GHz, 0.2 in transmission mode, 2.0 to 12.4 GHz, typically 0.2, 12.4 to 18 GHz.

Directivity 30 dB, 2.0 to 12.4 GHz, 18 dB, 12.4 to 18 GHz.

Reference plane extension 0 to 15 CM for reflection, 0 to 30 CM for transmission.

Connectors RF input, type N female, all other connectors APC-7.

Remote programming ground closure.

14. Calibration interval. After calibration, the equipment shall meet each performance requirement within the tolerances specified for a period of at least 9 months, based on an operating time of 1500 hours.

Preservation, packaging, labeling and marking. The preservation, packaging, packing labeling and marking shall be as specified in the contract or order.

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Certification. The contractor shall certify that the product offered meets the salient characteristics of this description and that the product conforms to the producer's own drawings, specifications, standards and quality assurance practices and is the same product offered for sale in the commercial marketplace. Small businesses that cannot offer products with established commercial market acceptability, may compete by submission of bid samples. The Government reserves the rights to require proof of such conformance before first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Known acceptable products. The following product(s) are known to have previously met the requirements of this CID and need not resubmit bid samples or certification. All other manufacturers must meet the requirements of this CID and submit the bid testing and certification as required.

Manufacturer	Model	NSN
Hewlett-Packard	8743B	6625-01-118-4683

Custodian:
Air Force - 99

Preparing activity:
Air Force - 82

(Project 6625-F884)

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